3. The Worker As the Controlling Component

Introduction

In the previous chapter, the characteristics of the worker as part of the worker/machine system were discussed at some length. In this chapter, the role of the worker as the controlling component in the worker/machine system is discussed.

Within the worker/machine system, the worker can act as a controller. The worker senses where a modification in the operation is necessary. The worker determines the extent of the modification that is necessary and makes the adjustments that are required to modify the action. In this manner, the worker acts to control the operation of the mechanical or electrical system being utilized. The worker observes what is happening in the work environment, either by direct observation of the process or by observation of the displays that indicate the status of the process. The worker uses visual sense to identify the need for system adjustment. Also, the worker listens to hear what is going on while the process is under way. Various changes in system operation result in auditory stimuli indicating progress is as expected or that the system is not progressing satisfactorily. A common example of this is the driver of a car who hears a strange noise while driving down the road. This indicates to the driver that something mechanical is wrong with the automobile and that service may be necessary. In this manner, the driver uses his auditory sense to identify the need for system adjustment.

Finally, the worker can use other senses in particular situations to identify the need for system adjustment. These senses are termed the "somesthetic senses" and include the senses of touch, taste, smell, and temperature. Though these senses are seldom used as primary sensors in the work environment, they often come into play. For example, often the first indication of a problem is the fact that the sense of smell indicates an odor that is not normal in the environment.

After having used the senses to determine the fact that something is awry in the system, the worker must determine what adjustment, if any, should be made in the system. To accomplish this, the worker must use mental capacity. The worker decides on the action that is necessary based upon previous training and experience relating the observed sense stimulus to its potential cause.

Finally, to implement the adjustment, the worker communicates the change to the system. This communication may be accomplished by using language or signals to communicate to another worker who in turn makes the actual changes

in the system, or the communication may be made directly to the system by adjusting the equipment controls.

The feedback loop is closed by providing information back to the worker's senses regarding the results of the changes or adjustments made in the system. From this information, the worker decides on the necessity for further adjustment to the system.

The Worker as a Sensor--Visual

Humans employ two different types of vision to determine or sense changes in the environment. The eye contains two types of sensing mechanisms. These mechanisms are termed "cones" and "rods." The cones are sensitive to the wave length of light; thus, cones are sensitive to color differences. On the other hand, the rods are sensitive to the amount of light or brightness.

Colorblindness is a lack of the ability of the cones to function properly. Colorblindness is seldom total. Most often, colorblindness is related to the inability of the human to distinguish various hues of red. green, and blue.

The ability to distinguish between colors is also affected by the intensity of light in the environment. You may have noticed on a dark, dreary morning how difficult it is to select the right color of socks from your drawer. This problem is often identified only when you get into the light and notice that one sock is brown and the other is black. The reason for this is that in dark areas vision depends upon the rods that are not color sensitive, and thus color discrimination is limited.

Visual Discrimination. There are two types of discrimination that can be employed by the human. The first is relative discrimination where two objects are viewed and their difference compared. The second is absolute discrimination where comparative information is not available and only one object is being viewed. Relative discrimination allows for a much greater number of differences to be identified by the individual than absolute discrimination does.

Some research has been conducted to identify the number of discriminations that can be made in a relative mode and an absolute mode (Mobray and Gebhard, 1958). The results of this research indicated that for relative discrimination, the following is found:

Relative Discrimination

1. Brightness--570 discriminable intensities were identified
2. Hues--128 discriminable hues at medium intensities were identified.
3. White light--375 discriminable interruption rates between 1 and 45 seconds were identified.

Absolute Discrimination

1. Brightness--3 to 5 discriminable intensities were identified.

2. Hues--12 to 15 discriminable hues were identified.

3. White light--5 to 6 discriminable rates were identified.

A similar situation was found for discrimination in the auditory senses. In a relative mode, 325 levels of loudness were discriminable. When an absolute discrimination was used, only 3 to 5 levels of loudness were discriminable by the subjects. In terms of pure tones, 1800 discriminable tones were identified in a relative mode while only 4 to 5 discriminable tones were identified in an absolute sense.

There are a number of factors that affect human visual sensitivity and ability to discriminate between various stimuli. Among the major factors are:

1. Illumination--The amount of illumination that is present and measured in footcandles is a major factor in determining the ability to discriminate. The amount of illumination necessary depends upon the detail of the work required. Where fine motor skills are required to assemble or produce the desired product, such as in mechanical drafting or inspection of finished product, the amount of illumination present must be high. On the other hand, if the worker is digging a ditch, illumination need not be nearly as intense.

2. Glare--The amount of glare in the work area can affect the ability of the worker to discriminate. The eyes are adjusted to one level of light, while the glare is at a different level of light. This glare adversely affects the visual acuity of the worker and makes discrimination difficult if not impossible.

3. Contrast--The ratio of difference between the reflectance of an object's background to the reflectance of the object is termed "contrast." Often the term used is "luminance contrast." This contrast can be calculated by the formula below.

$$\text{Contrast} = \frac{R_1 - R_2}{R_1\ (100)}$$

where
R_1 = the higher reflectance object
R_2 = the lower reflectance object

4. Changing light--As the light changes from bright to dark, the eye must adjust to allow for more light to enter. Time is required for the adjustment to occur. During this adjustment period, the ability to discriminate visually is greatly reduced.

5. Distance--The size of an object relative to the distance from which it is viewed is important in the ability of an individual to discriminate. Obviously the size must increase as the distance increases. However, this obvious fact is often overlooked in the preparation of materials for viewing by large audiences at a distance.

6. Angle of view--The angle of view can distort the visual stimulus being viewed. This is particularly the case when viewing through glass where a parallax distortion occurs when viewing at an angle. A 90° angle is best for viewing accuracy.

7. Time to view--As the time increases, the discrimination increases to a point of diminishing returns. Time is also dependent upon other variables, such as the illumination, the distance of viewing, and the angle of view.

8. Movement--If the object being viewed is moving, the acuity of vision is not as great. The faster the movement of the object, the lower the visual acuity and the ability to discriminate.

9. Distractions--Distractions in the viewing area can cause a problem sometimes termed the "signal-to-noise ratio." The object being viewed is the signal, and any distractions are noise. If the signal-to-noise ratio is low, then the distraction stimuli will tend to mask the target stimuli that is being viewed. The viewer may then find it impossible to keep his attention directed to the signal object. This often happens when the viewer must split his attention between a number of displays. Errors are more likely, and identification of key situations is slower.

The Worker as a Sensor--Auditory

The reader will recall from previous discussions related to noise that the sound intensity or sound-pressure level is measured in terms of the decibel. The decibel is a logarithmic ratio of sound intensity or sound-pressure level with a fixed standard base. Sound frequency, measured in hertz (cycles/second) is related to the tones that are heard. The human ear is sensitive in the range of 20 to 20,000 hertz.

Since human reaction to sound is a function of both pressure level and frequency, some research has been done to determine equal loudness curves for varying frequency and intensity as determined by human respondents. The phone is basically a subjective equality of different sounds with varying frequency and intensity. On the other hand, the sone is a measure of the relative subjective loudness of sounds. If a sound is twice as loud, then the sones are doubled.

Often in a work environment one sound will reduce the sensitivity of the ears to another sound. This effect is termed "masking." Masking can cause the threshold of audibility of the sound to be raised. The masking effect depends on the frequency and intensity of the two sounds; that is, the masking effect depends on the frequency of the signal versus the frequency of the noise and the intensity of the signal versus the intensity of the noise.

Again, the signal-to-noise ratio is applicable. The noise acts in the environment to mask the signal. In the case where noise is a problem the signal can be increased in strength, or the noise level can be lowered. A general rule is that the signal should be increased in strength to a point one-half the way between the noise level and 100 decibels. At this point, the sound should be audible to most individuals. On the other hand, the noise can be decreased through the use of earmuffs that select out certain tones which act to mask the signal to be heard. Where electronic audio systems are present, electrical filters can be used to mask out noise or static that occurs, thus producing a clearer signal. Of course, there are also the many techniques that are available for noise reduction in the environment. These have been discussed previously in the section on the control of noise.

Auditory displays of a number of different types exist. First, there is voice communication where one individual communicates to another certain commands to be carried out. A second type of auditory display relates to the need for the operator to make adjustments to the system. A changing tone frequency can be used to indicate when adjustments are necessary and the results of these adjustments. The sound level itself can be varied to indicate the effect of adjustments to the system. Finally, an intermittent sound can be used as an auditory display. This type of auditory display is most commonly found where a warning signal is given to indicate that adjustments to the system are needed quickly.

The Worker as a Sensor--Tactual

The most common somesthetic sense that the worker is most likely to use in the work environment is the tactual or touch sense. In some cases it is necessary that the worker use his touch sense to determine system status and to exercise control. This may happen when it is necessary for an individual worker to reach and grasp a given object to be used while at the same time keeping his eyes fixed on another object.

Using the tactual sense, the worker can discriminate in a number of different ways. Using the tactual sense, man can discriminate between the sizes and shapes of objects. The texture of an object can also be determined by touch. In addition, the relative distance or location can often be determined by touch and positioning of the hands with the necessity of viewing the object.

Some of the other somesthetic senses that often come into play in the work environment are the sense of temperature, pain, vibration, and pressure. Although these senses are seldom used as a primary method for obtaining information concerning the state of the system, they can come into play when something goes wrong by providing the worker with an additional warning mechanism.

Displays

The purpose of a display in the workplace is to help the worker to sense changes in the state of the system where such changes might not be of a magnitude that normally would be sensed. There are basically three kinds of

displays that are commonly used to assist the worker in sensing changes in the system. These types of displays are visual displays, auditory displays, and tactual displays.

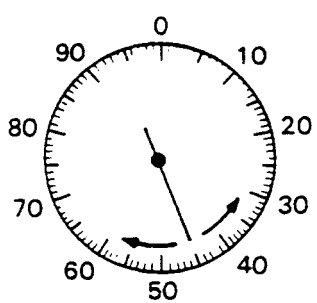
Four basic types of visual displays are common. The type of visual display is based on the information obtained from the display. The types are:

1. Quantitative displays
2. Qualitative displays
3. Representational displays
4. Status displays

A quantitative display tells the operator or worker how much. It may measure a static variable, as a yardstick does; or it may measure a dynamic variable, as a gauge does. A common example of a quantitative display that measures a dynamic variable is the speedometer on an automobile. This instrument gives a readout to the operator concerning the actual velocity of the automobile as well as indicating the rate of change or acceleration of the automobile by the direction of movement of the pointer.

Figure 7.3.1

Quantitative display.

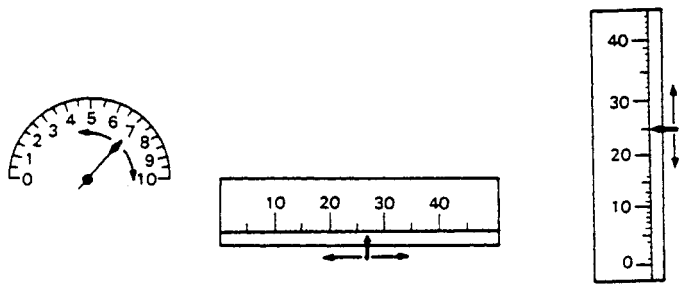


There are various types of display designs that are used for quantitative displays. The display may be presented as a circular scale or as a semi-circular scale. In some cases, vertical and horizontal scales might be used for the display. Finally, a digital display can be used. Research favors the use of a circular or semi-circular scale for display purposes. However, there are circumstances in which vertical and horizontal scales may have advantages and should be used. Digital scales have the advantage of both accuracy and time to read. However, the rate of change for a digital scale is not so easily determined as for scales of the other type.

Another concern is the type of display operation to be utilized. There are basically two types that can be used--the fixed-scale-moving-pointer display and the fixed-pointer-moving-scale display. In general, research has

Figure 7.3.2

Scales.



shown that the fixed-scale-moving-pointer display is preferable in most cases. However, again there may be instances where the moving-scale-fixed-pointer provides the operator with information in a better form.

In terms of the general design of a quantitative scale, it is important that the scale displays the smallest and largest values possible. The interval between readings depends on the accuracy that is necessary. If, for example, a display is to be read to the nearest 10 feet, then 10 feet would be the smallest unit of measurement. The scale would be constructed in such a manner that a given length of the scale would be equal to 10 feet. Research has indicated that the difference between scale units should be between .05 and .07 inches to assure acceptable accuracy in reading the scale. One mustconsider the conditions of use to determine the actual size of the numbers and the scale intervals to be used. Obviously if the visual acuity is lessened because of poor illumination or distance, the size of the numbers and the scale interval must be greater.

Scale markers can be used to indicate the units to be read. In general, the easiest progressions used are in terms of 1's, 10's, 100's, etc. With these types of scales, intermediate markers are generally shown at the 5, 15, 25 levels. (See Figure 7.3.2.) In some cases, individual markers are shown between the intermediate marks. The preference for scales of this type is generally a matter of conditioning. Workers have become accustomed to reading scales of this type and need very little practice to read similar scales. In any case, the numbering should always progress from left to right, similar to the direction of reading.

The pointer should be close to the scale to avoid parallax distortion. The end of the pointer should just meet the smallest marker but should not overlap. It is generally recommended that the tip angle of the pointer be approximately 20°.

Qualitative Visual Displays

Qualitative visual displays are generally used for a relative determination of range or trend. One type of qualitative display can be obtained by color coding parts of a quantitative display. For example, the danger areas of a quantitative display may be color coded red while the normal operating areas are color coded green. Caution areas or areas immediately preceding the danger area can be color coded yellow.

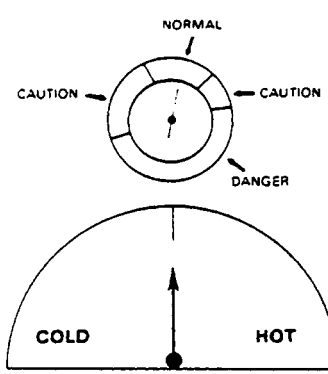
An example of a qualitative type visual display is the temperature display found in most automobiles. Normally, temperature is no longer shown on such a display, and only hot and cold areas of the gauge are indicated. The pointer will move between the cold and hot areas. The driver need not be concerned about the particular temperature, only that the pointer is somewhere in the middle area of the gauge.

Qualitative displays have the advantage of providing the information that is necessary for the operator to take the required action quickly with very little chance of error. In the quantitative display, a great deal of information can be provided to the worker. However, it is not always

necessary that the operator has such detailed information. In fact, research has shown that the addition of qualitative coding to a quantitative scale materially decreases the time required for recognition of certain key conditions. (Kurke, 1956.)

Figure 7.3.3

Qualitative displays.



Status Indicators. Status indicators are generally used for discrete situations. Often a dichotomous output is obtained, such as on-off, hot-cool, charge-discharge, etc. Status indicators can be either labeled switches, such as are found in a circuit box, or signal lights that indicate the need for action.

Figure 7.3.4

Representational displays.

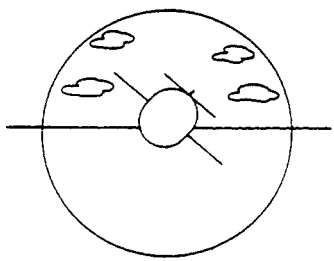
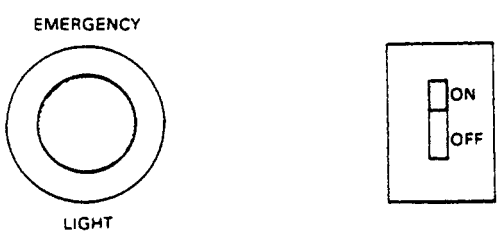


Figure 7.3.5

Status indicators.



A special case of the status indicator is the auditory alarm. This alarm is the most common use of the auditory signal. Often an auditory alarm is used in conjunction with a visual signal. The auditory alarm is more likely to be noticed than the visual signal; however, the visual signal acts as a reinforcement of the auditory alarm to indicate the point where action is needed.

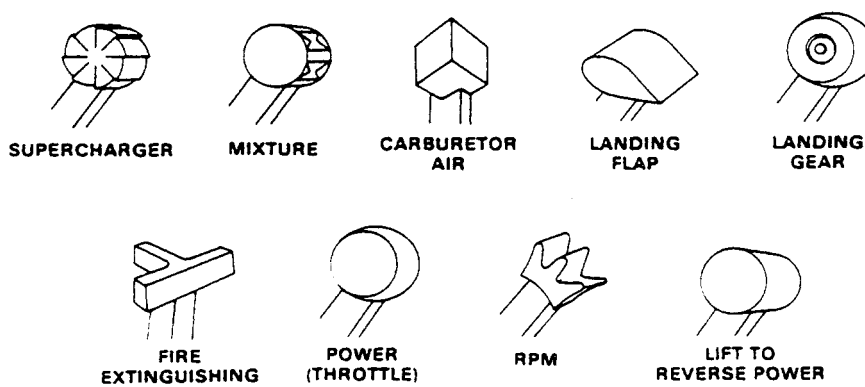
Auditory Range Displays. Auditory range displays are of two types. The intensity of the tone can be used to indicate the status of the system relative to the desired status by changes in the loudness or intensity of the tone. In another type of auditory display, different tones or patterns can be used to provide additional information, such as the status of the system being to the left or the right of the desired course. A constant signal can indicate that the system is on course. This type of system is commonly used for guidance in airplanes.

Tactual Display. Generally speaking, a tactual display is used as a display of information concerning the location of a control rather than system status. Control knobs are coded as to the function of the knob, either by shape or the location of the knob in question. Texture is another method of

coding control knobs in order that the operator can continue to maintain visual contact with another target while determining if the proper knob is being used.

Figure 7.3.6

Shape coding.



Choosing the Type of Stimulus

In the design of a system for human control of a machine or a system of machines, one of the major problems encountered is the type of stimulus that should be provided to the operators. Should a visual signal be used or is an auditory signal preferred? There are certain general rules that can be applied to this determination.

In general, visual signals are preferred in the following cases:

1. Where relative values or quantitative values are desired.
2. Where the information is complex.
3. Where there is a need to refer to the information at a later point.
4. Where there are a number of displays that must be monitored.
5. Where the location of the object relative to another object is desired.
6. Where the display is representative of the actual situation.
7. Where noise limits the use of auditory signals.

On the other hand, an auditory signal is preferred in the following cases:

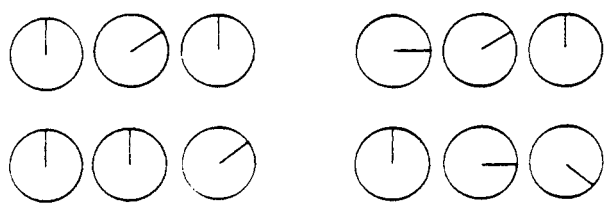
1. Where a warning is being given and action is desired.
2. Where too many visual stimuli are present.
3. Where information must be presented independent of head orientation.
4. Where visual acuity is limited.
5. By speech, where communication content may vary considerably.

Grouping of Visual Displays

Quite often it is necessary for the worker to monitor a number of displays. Multiple displays create the possibility of confusion to the worker unless organized in some logical fashion. The general method is to organize the display such that the pointer is in the same direction for normal operation. Thus, it is easy to pick out abnormal situations when they occur. The grouping tends to be viewed as a whole rather than individually, and variations in the pattern are easy to spot.

Figure 7.3.7

Pattern vs. non-pattern groups.



In terms of location of the display or groups of displays, the most significant displays--that is, those used most often--should be in the normal line of vision. Less important displays can be moved to an out-of-the-way position. Warning signals may allow some types of displays to be moved to an out-of-the-way position in which they need be read only if the warning signal goes off.

Summary

Within the worker/machine system, the worker often must act as a controller. In this position, it is necessary that information concerning the status of the system be presented to the worker. This information can be presented in such a way that the worker can use one of many senses to determine changes in the system. In general, the most common method for providing information to the worker is through visual displays. This type of display is normally either quantitative or qualitative in nature depending on the type of information required by the operator. Auditory displays are used in certain special conditions, particularly as warning devices where the worker's vision must be directed elsewhere. Generally, tactual displays are used to provide the worker with information concerning the appropriate control to use in a given situation when it is necessary that the worker maintain visual contact with another area.